

Project Proposal Preview

Request: I want to add a search bar in my website

1. Requirements

Add a search bar to the website that allows users to search content

Functional requirements

- Provide a text input with placeholder and submit button or enter key trigger
- Display search results in a dropdown or dedicated results page
- Highlight matched terms in results
- Show a message when no results are found
- Allow basic filtering by category if categories exist
- Support pagination if results exceed a threshold

Non-functional requirements

- Responsive layout for mobile and desktop
- Search results load within 200ms
- Accessible with ARIA labels and keyboard navigation
- Minimal impact on page load performance
- Use vanilla JS without external libraries
- Compatible with modern browsers

Edge cases

- Empty query string should prompt user or show all results
- Special characters in query should be escaped or handled gracefully
- Very long queries should be truncated or limited
- Network failure when fetching results should display error
- Duplicate results should be deduplicated
- Case sensitivity should be consistent

Assumptions

- Search data is available via a JSON file or API endpoint
- No authentication is required for searching
- Users are on modern browsers supporting ES6
- Search is performed client-side only
- The website uses plain HTML, CSS, and JS
- No server-side rendering is involved

2. Architecture

Implement a vanilla JS client-side search module that loads a static JSON data file, filters results on user input, and updates the DOM with a dropdown list or results page

Component	Responsibility	Type
SearchUI	Render input field, results container, handle user interactions (input, submit, pagination)	frontend

Component	Responsibility	Type
SearchLogic	Load data, filter based on query, highlight matches, deduplicate, handle edge cases	frontend
DataFetcher	Fetch the static JSON data file via fetch API and expose it to SearchLogic	integration

Data flow

- User focuses or types in the search input
- Input event (debounced) triggers SearchLogic to ensure data is loaded via DataFetcher
- SearchLogic filters the dataset according to the query, applies case-insensitive matching, highlights terms, and removes duplicates
- SearchLogic returns result set to SearchUI
- SearchUI renders results in a dropdown or navigates to a dedicated results page; shows 'no results' or error messages as needed

File structure

- index.html
- css/style.css
- js/search.js
- data/search-data.json

3. Tools

4. Cost Estimate (tools/services only)

Estimated monthly total: \$0

Note: These are third-party tool/service cost estimates only, not a cost of building the feature